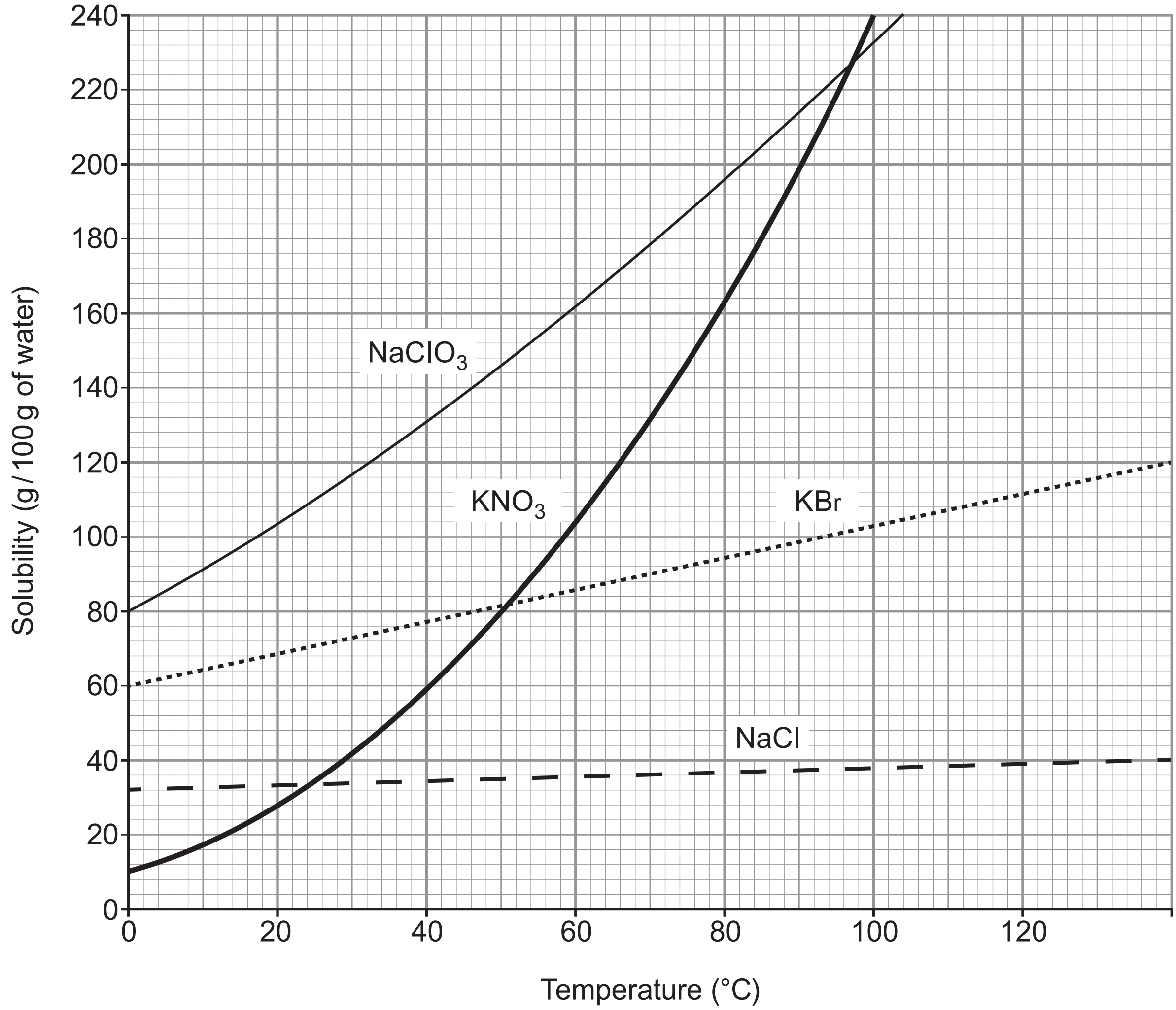
2. The grid below shows the solubility curves for four ionic compounds.



$NaClO_3$	sodium chlorate
KNO_3	potassium nitrate
KBr	potassium bromide
NaCl	sodium chloride

Examiner only

(a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

.... °C

(ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = ... g

(iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

....

....

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(b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

....

(ii) Explain how these ions are formed from their atoms. [2]

....

....

(c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

Balance the symbol equation for the reaction taking place. [1]

$$\boxed{}\,KNO_3 + Al(OH)_3 \longrightarrow Al(NO_3)_3 + \boxed{}\,KOH$$

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